

GPU Blosc: throughput / compression Pareto frontiers

NVIDIA L40 · nvCOMP 5.3 · driver 580.126.20 · sweep 2026-09-06 UTC

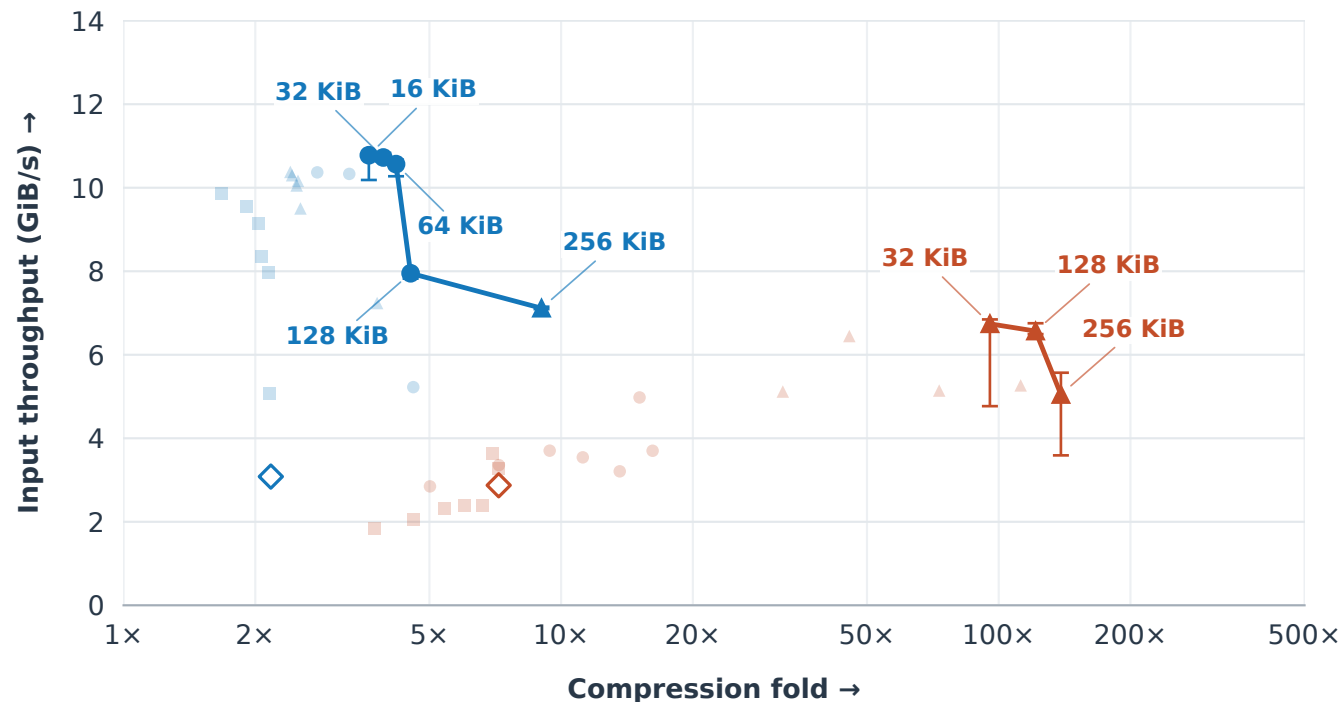
Per-codec frontier across block sizes + all shuffle modes. Higher and farther right is better.

— LZ4 — Zstd ■ no shuffle ● byte shuffle ▲ bitshuffle ◇ raw codec control

Labels: Blosc block size · bars: min-max of 5 runs · faint points: dominated candidates

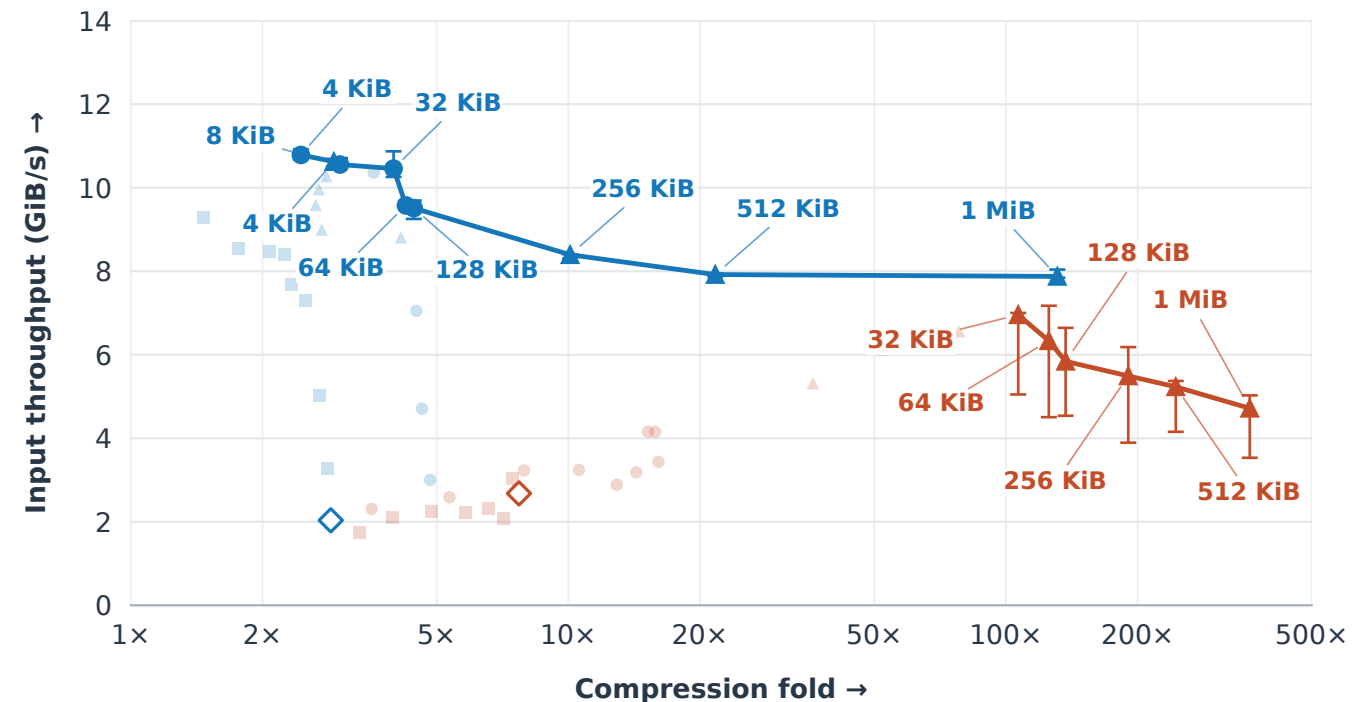
Repetitive XOR · 256 KiB Zarr chunks

log ratio axis



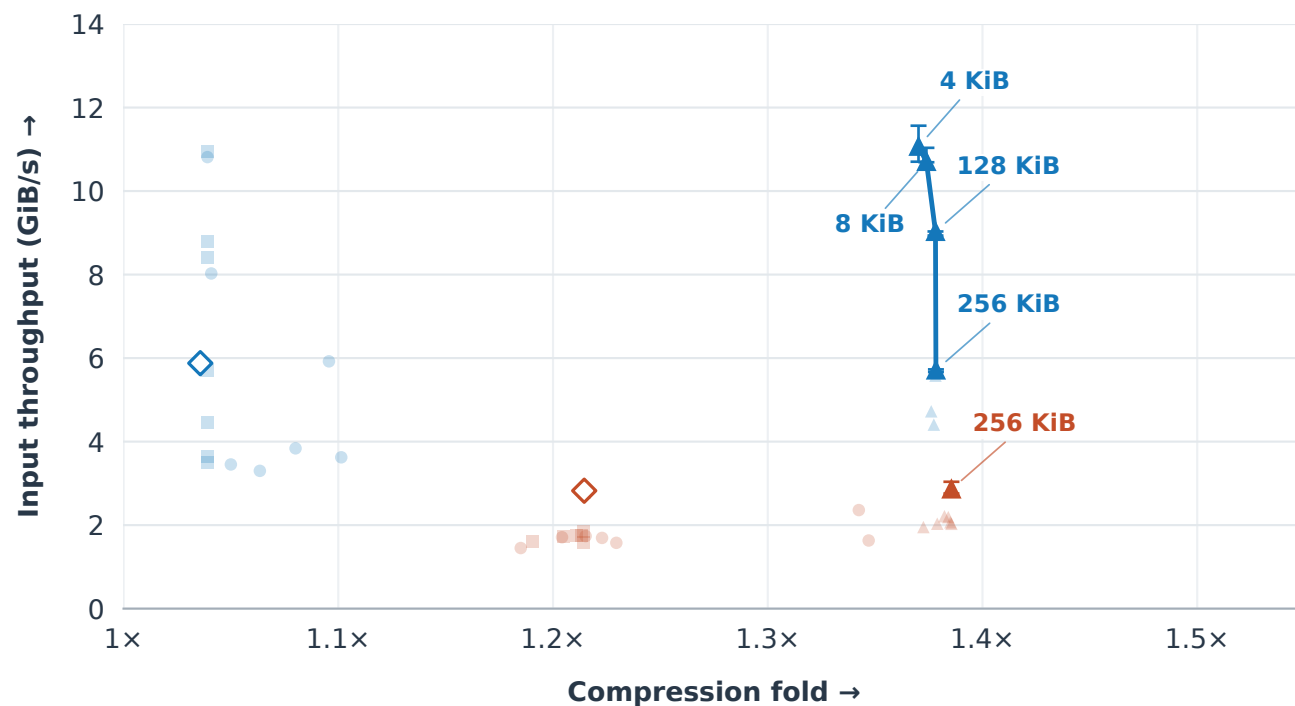
Repetitive XOR · 1 MiB Zarr chunks

log ratio axis



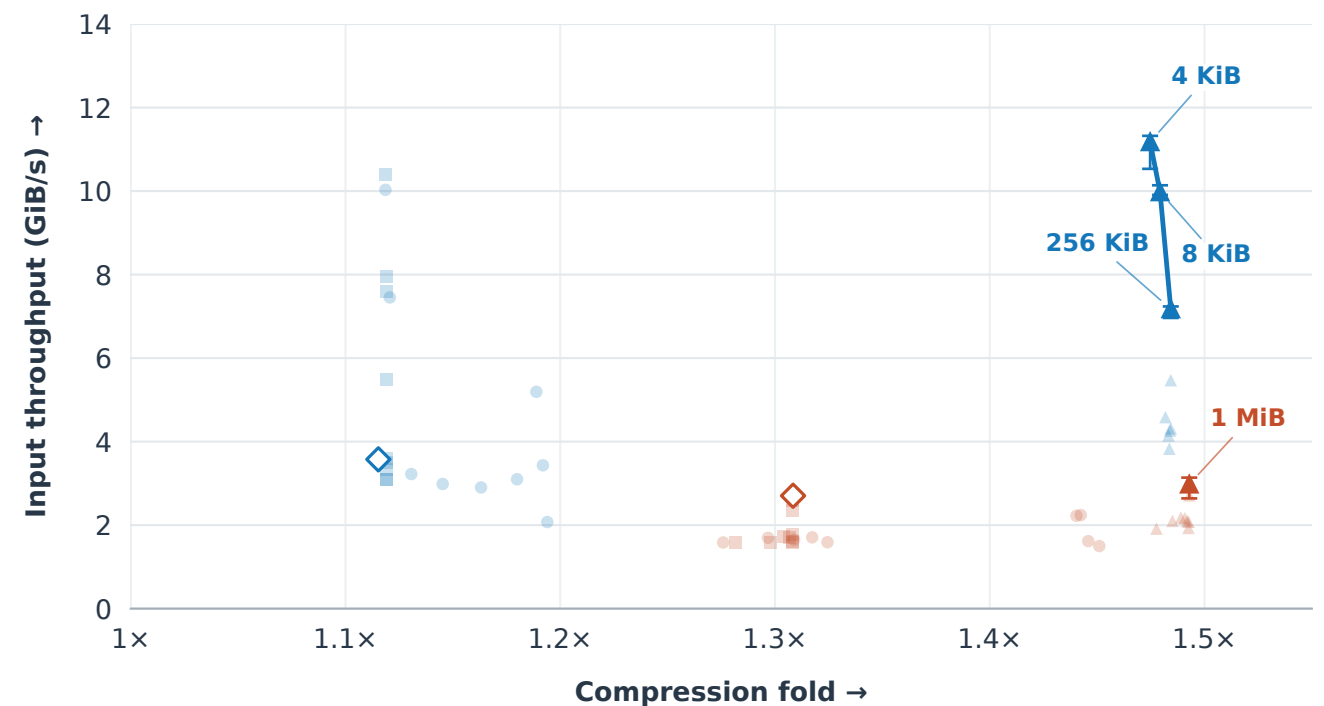
12-bit random · 256 KiB Zarr chunks

linear ratio axis



12-bit random · 1 MiB Zarr chunks

linear ratio axis



Frontiers compare tested Blosc settings within each codec; both median throughput and compression fold are maximized.

Fold = padded chunk bytes / encoded bytes. Synthetic input and discard sink; GPU memory is not an objective here.

Lines join measured settings. Bars show observed spread, not confidence intervals.

All measured candidates shown. Exact-median frontier membership can be sensitive to small differences.